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Yamagoe

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(54) **BOOM CONTROL SYSTEM OF WORK MACHINE**

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E02F 9/22 (2006.01)

(52) **U.S. Cl.**
CPC ***F15B 11/044*** (2013.01); ***E02F 9/2203*** (2013.01); ***E02F 9/2207*** (2013.01); *F15B 2211/3057* (2013.01); *F15B 2211/3127* (2013.01)

(58) **Field of Classification Search**
CPC E02F 9/2203; E02F 9/2207; F15B 11/044; F15B 2211/3057; F15B 2211/3127
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,138,838 A * 8/1992 Crosser F15B 21/087 91/463
5,598,648 A 2/1997 Moriya et al.
2003/0230082 A1 12/2003 Wook Kim
2004/0011192 A1* 1/2004 Frediani F15B 11/044 91/465

(Continued)

FOREIGN PATENT DOCUMENTS

CN 101922162 A 12/2010
CN 104302930 A 1/2015

(Continued)

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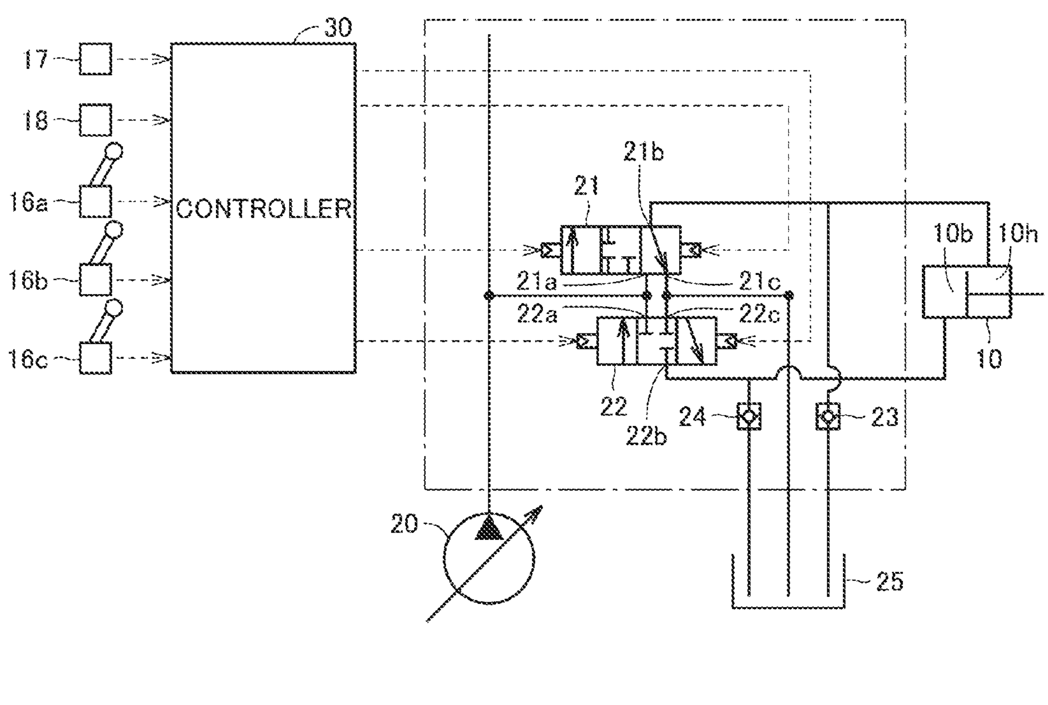
(57) **ABSTRACT**

A boom cylinder drives a boom, and includes a head side oil chamber and a bottom side oil chamber. A first valve includes a first opening for discharging oil in the head side oil chamber to an oil tank, and supplies oil from a hydraulic pump to the head side oil chamber. A second valve includes a second opening for discharging oil in the bottom side oil chamber to the oil tank, and supplies oil from the hydraulic pump to the bottom side oil chamber. A controller individually controls an opening degree of the first opening and an opening degree of the second opening, during work including manipulation of the boom.

10 Claims, 7 Drawing Sheets



(56) **References Cited**

U.S. PATENT DOCUMENTS

2010/0313557	A1	12/2010	Kim	
2012/0163949	A1	6/2012	Okano et al.	
2015/0107234	A1	4/2015	Ueda et al.	
2016/0333551	A1	11/2016	Joung et al.	
2018/0128292	A1*	5/2018	Wang	F15B 11/006
2019/0024677	A1*	1/2019	Rannow	E02F 9/2203

FOREIGN PATENT DOCUMENTS

CN	110144987	A		8/2019	
GB	2298291	A	*	8/1996	E02F 9/2025
JP	S52-038101	U		3/1977	
JP	H03-066838	A		3/1991	
JP	H10-168950	A		6/1998	
JP	H11-158942	A		6/1999	
JP	2000-204601	A		7/2000	
JP	2011-002092	A		1/2011	
JP	2011-052766	A		3/2011	
JP	2011-236562	A		11/2011	
JP	2019-210670	A		12/2019	
JP	2020-153506	A		9/2020	
JP	2021-038792	A		3/2021	
KR	10-2015-0030328	A		3/2015	

* cited by examiner

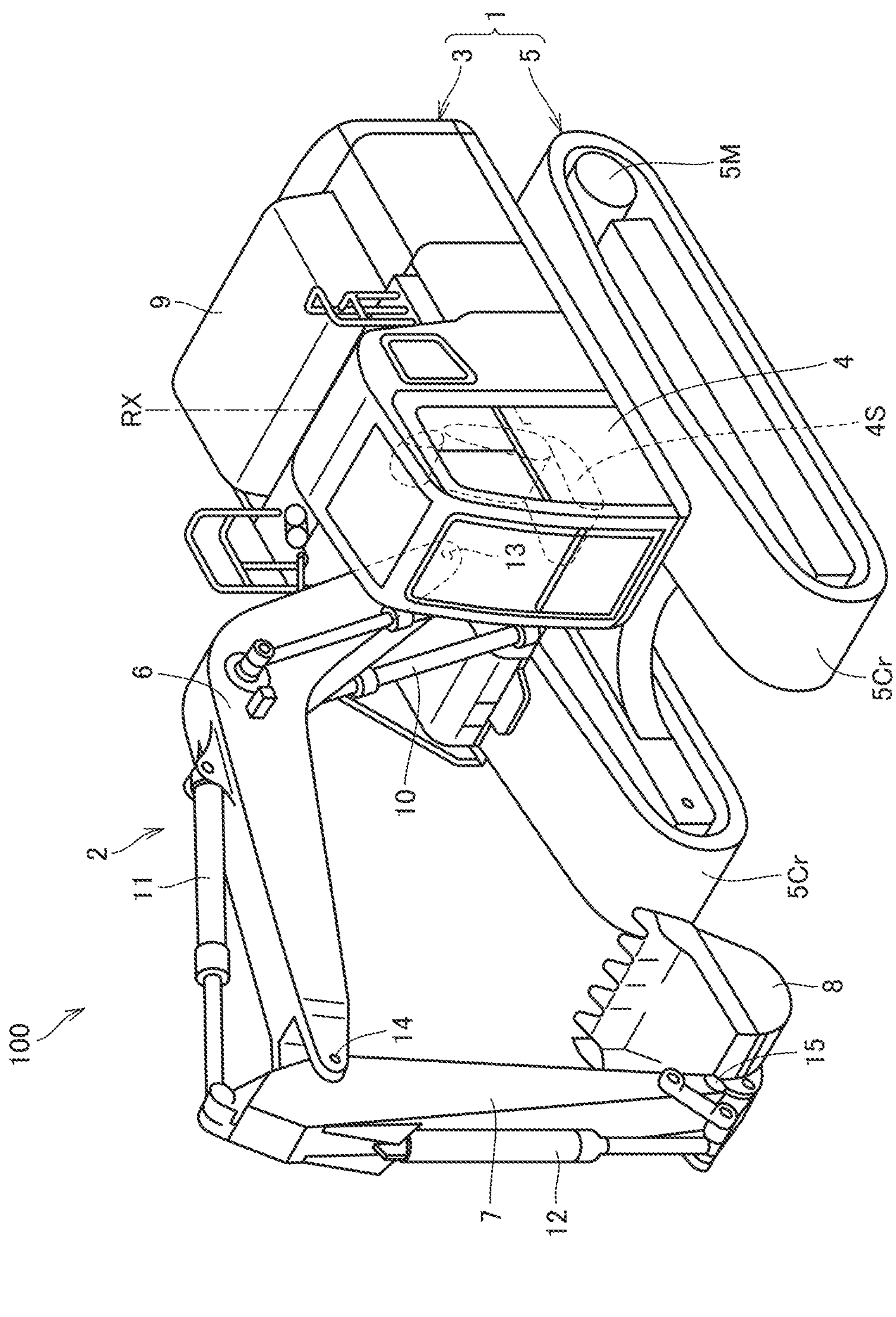


FIG.1

FIG.2

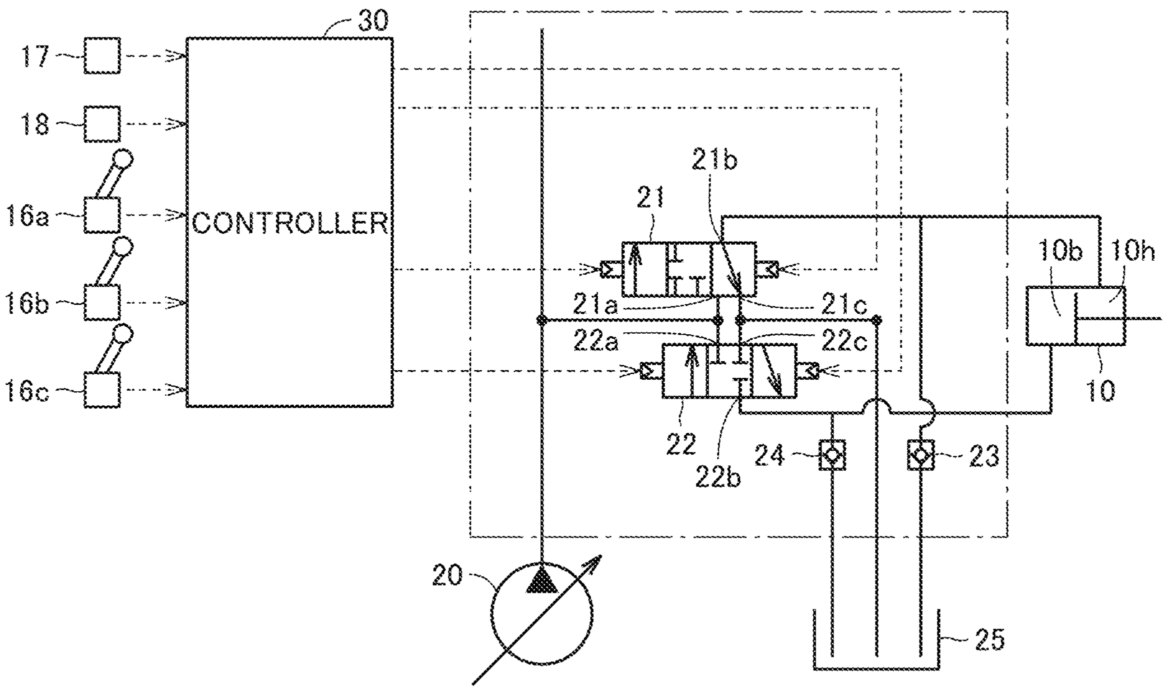

30
17
18
16a
16b
16c
CONTROLLER
21
21b
21a
21c
22a
22c
10b
10h
10
22
22b
24
23
20
25

FIG.3

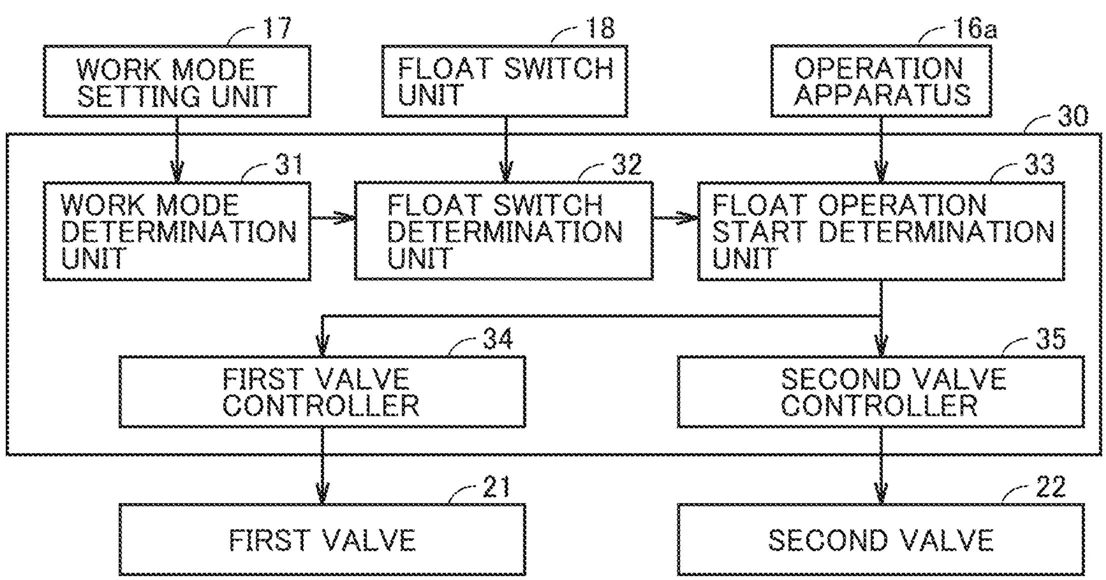

17
WORK MODE SETTING UNIT
18
FLOAT SWITCH UNIT
16a
OPERATION APPARATUS
30
31
WORK MODE DETERMINATION UNIT
32
FLOAT SWITCH DETERMINATION UNIT
33
FLOAT OPERATION START DETERMINATION UNIT
34
FIRST VALVE CONTROLLER
35
SECOND VALVE CONTROLLER
21
FIRST VALVE
22
SECOND VALVE

FIG.4

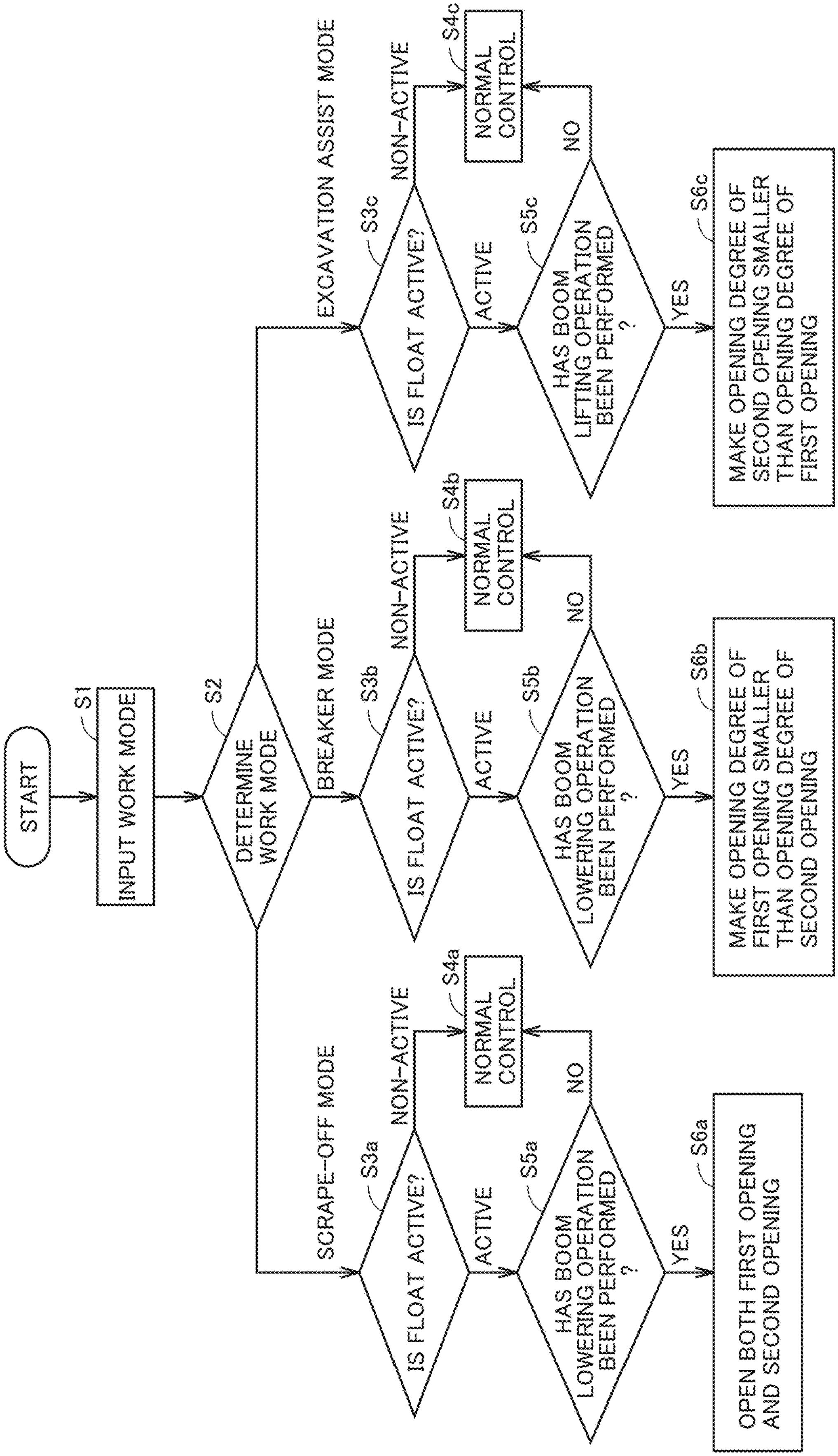

START
S1
INPUT WORK MODE
S2
DETERMINE WORK MODE
SCRAPE-OFF MODE
BREAKER MODE
EXCAVATION ASSIST MODE
S3a
IS FLOAT ACTIVE?
S3b
IS FLOAT ACTIVE?
S3c
IS FLOAT ACTIVE?
NON-ACTIVE
ACTIVE
S4a
NORMAL CONTROL
S4b
NORMAL CONTROL
S4c
NORMAL CONTROL
S5a
HAS BOOM LOWERING OPERATION BEEN PERFORMED ?
S5b
HAS BOOM LOWERING OPERATION BEEN PERFORMED ?
S5c
HAS BOOM LIFTING OPERATION BEEN PERFORMED ?
YES
NO
S6a
OPEN BOTH FIRST OPENING AND SECOND OPENING
S6b
MAKE OPENING DEGREE OF FIRST OPENING SMALLER THAN OPENING DEGREE OF SECOND OPENING
S6c
MAKE OPENING DEGREE OF SECOND OPENING SMALLER THAN OPENING DEGREE OF FIRST OPENING

FIG.5

<SCRAPE-OFF>

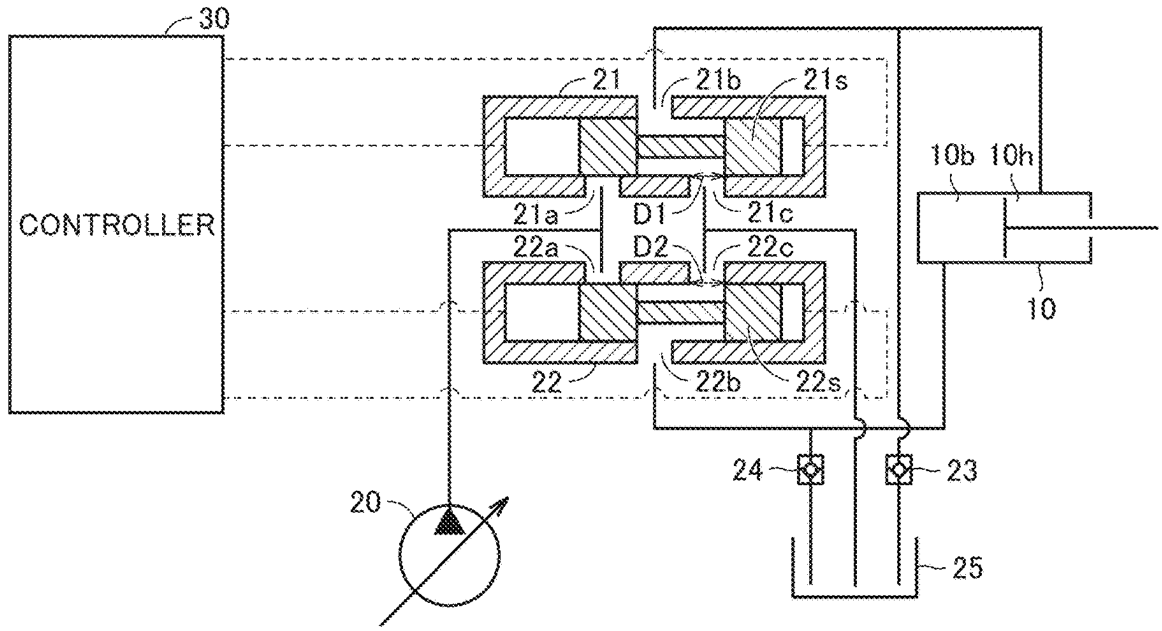


FIG.6

<BREAKER>

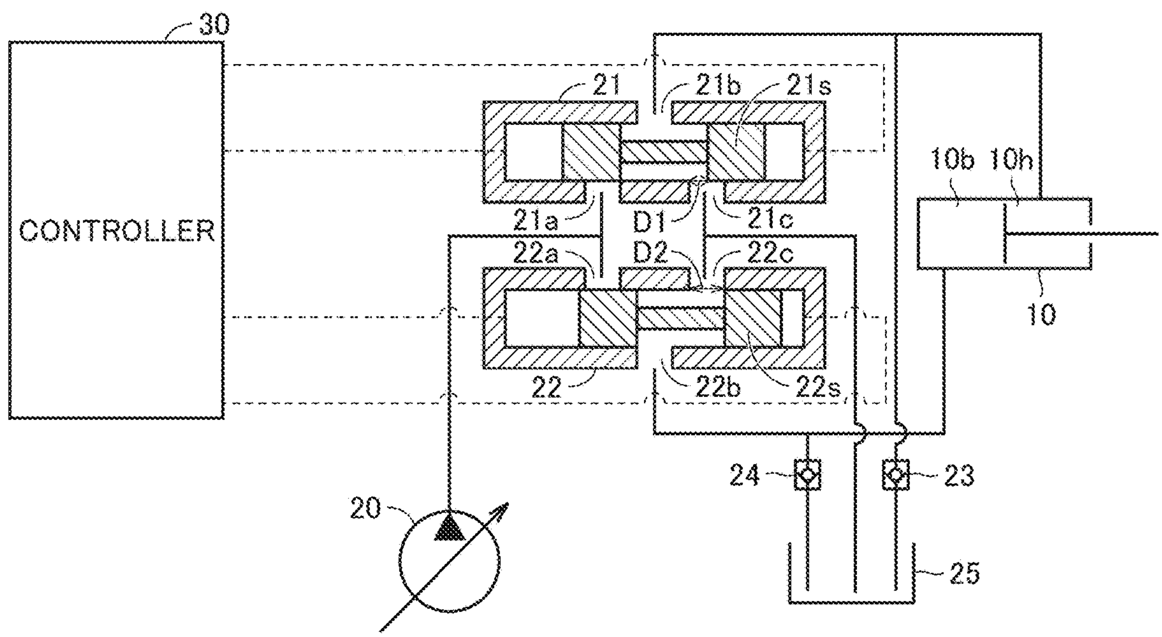
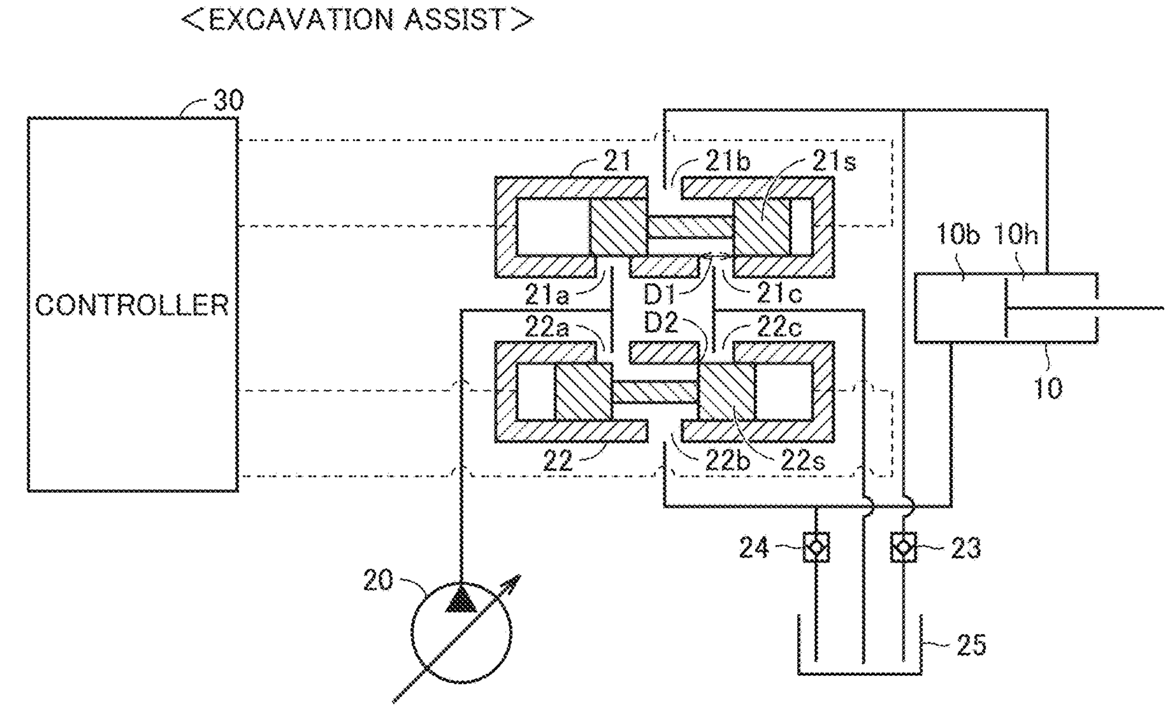
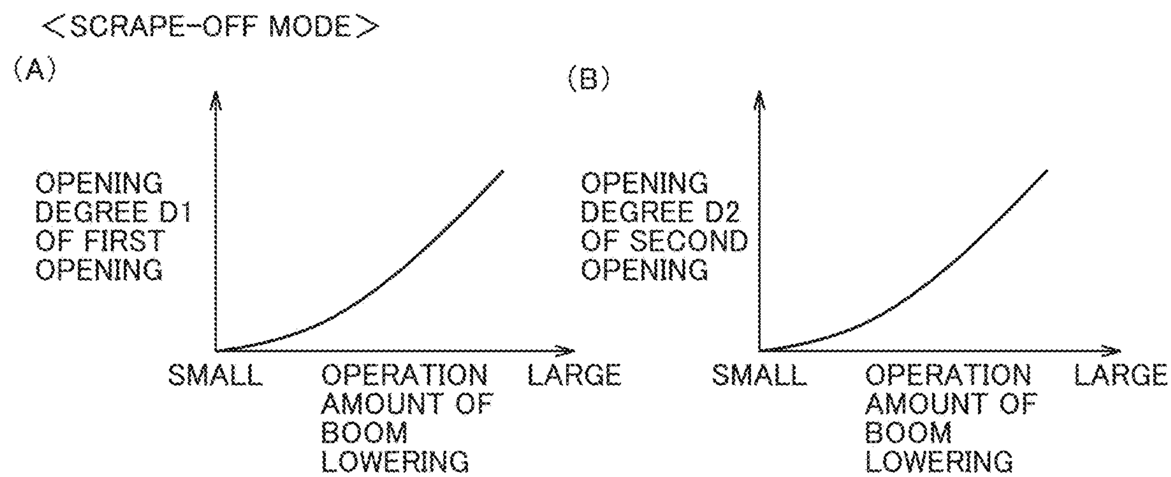


FIG.7



<EXCAVATION ASSIST>
30
CONTROLLER
21
21b
21s
21a
D1
21c
22a
D2
22c
22
22b
22s
10b
10h
10
24
23
20
25

FIG.8



<SCRAPE-OFF MODE>
(A)
OPENING DEGREE D1 OF FIRST OPENING
SMALL
OPERATION AMOUNT OF BOOM LOWERING
LARGE
(B)
OPENING DEGREE D2 OF SECOND OPENING
SMALL
OPERATION AMOUNT OF BOOM LOWERING
LARGE

FIG.9

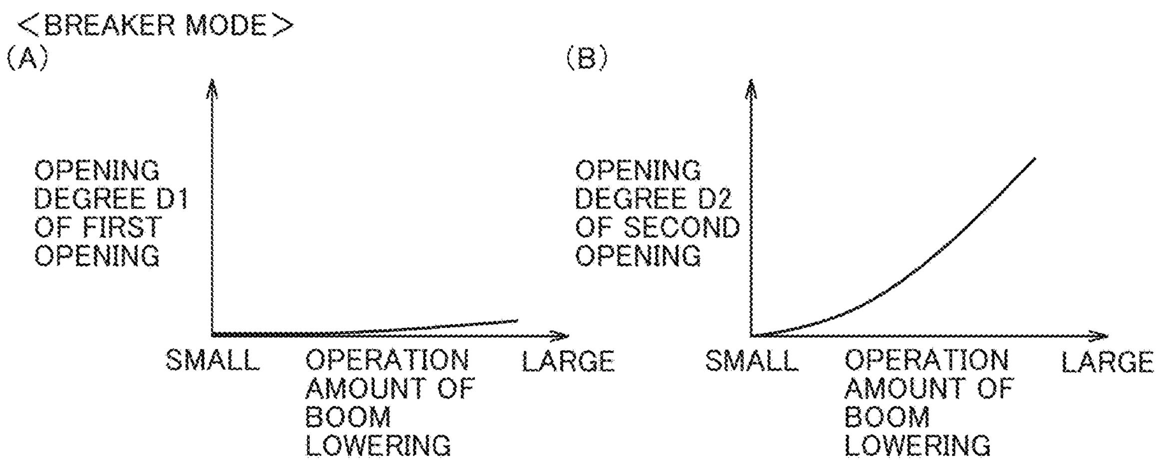

<BREAKER MODE>
(A)
OPENING DEGREE D1 OF FIRST OPENING
SMALL
OPERATION AMOUNT OF BOOM LOWERING
LARGE
(B)
OPENING DEGREE D2 OF SECOND OPENING
SMALL
OPERATION AMOUNT OF BOOM LOWERING
LARGE

FIG.10

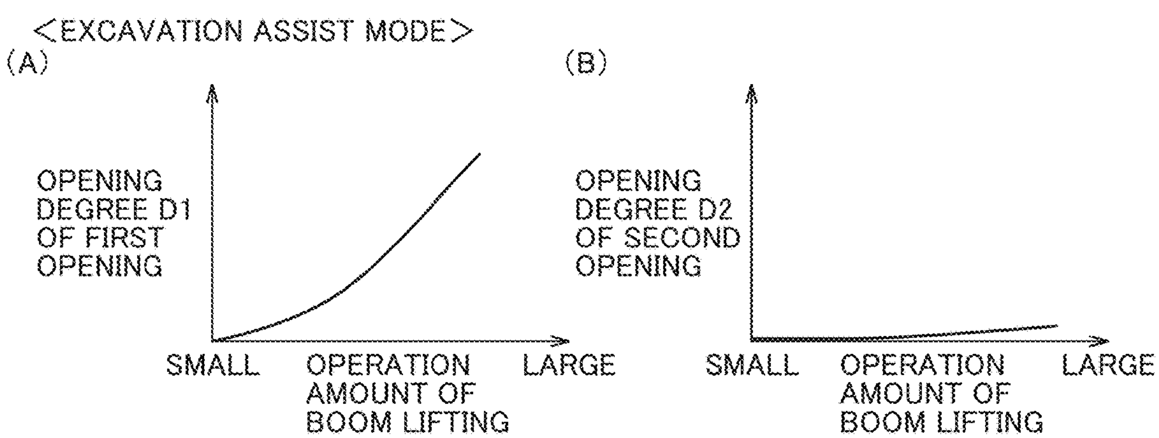
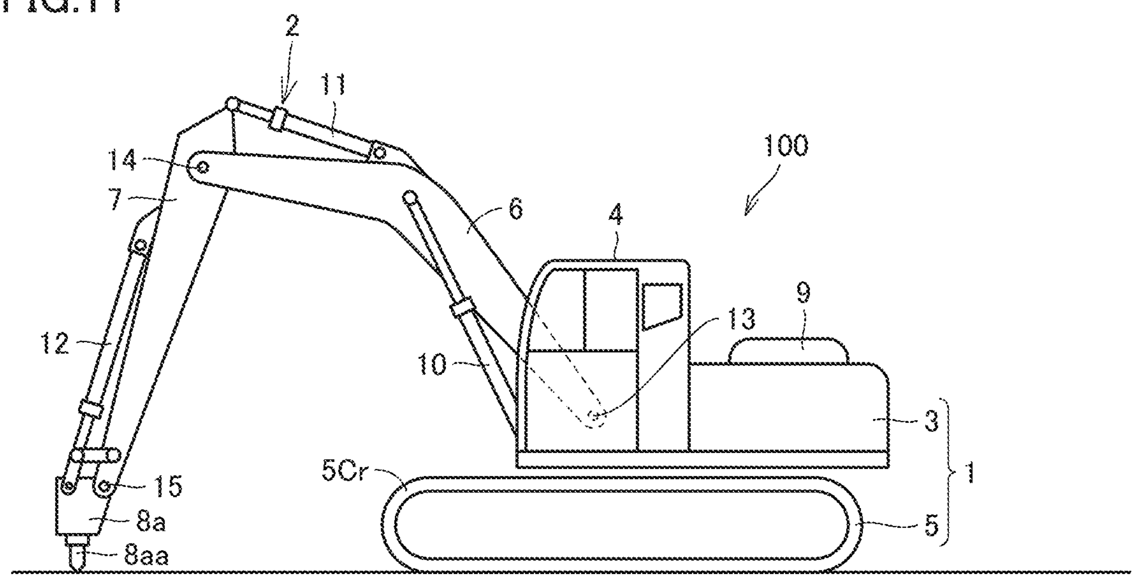

<EXCAVATION ASSIST MODE>
(A)
OPENING DEGREE D1 OF FIRST OPENING
SMALL
OPERATION AMOUNT OF BOOM LIFTING
LARGE
(B)
OPENING DEGREE D2 OF SECOND OPENING
SMALL
OPERATION AMOUNT OF BOOM LIFTING
LARGE

FIG.11



BOOM CONTROL SYSTEM OF WORK MACHINE

TECHNICAL FIELD

The present disclosure relates a boom control system of a work machine.

BACKGROUND ART

Japanese Patent Laying-Open No. H3-66838 (PTL 1) for example discloses a work machine such as hydraulic excavator having a boom float function. The boom float function refers to a function of enabling a boom to swing freely by having a head side oil chamber and a bottom side oil chamber of a boom cylinder communicating with an oil tank, without discharging hydraulic oil from a hydraulic pump to the boom cylinder.

CITATION LIST

Patent Literature

PTL 1: Japanese Patent Laying-Open No. H3-66838

SUMMARY OF INVENTION

Technical Problem

Work machines are utilized for various uses. In recent years, therefore, there has been a demand for the boom float function that can be applied to various uses.

In view of the above, an object of the present disclosure is to provide a boom control system of a work machine for which the boom float function is adjustable depending on a current work mode.

Solution to Problem

A boom control system of a work machine of the present disclosure includes a boom, a boom cylinder, a hydraulic pump, an oil tank, a first valve, and a second valve. The boom cylinder drives the boom and includes a head side oil chamber and a bottom side oil chamber. The first valve supplies oil from the hydraulic pump to the head side oil chamber and discharges the oil to the oil tank. The second valve supplies oil from the hydraulic pump to the bottom side oil chamber and discharges the oil to the oil tank. The first valve includes a first opening for discharging the oil in the head side oil chamber to the oil tank. The second valve includes a second opening for discharging the oil in the bottom side oil chamber to the oil tank. The boom control system of the work machine further includes a controller that individually controls an opening degree of the first opening and an opening degree of the second opening, during work including manipulation of the boom.

Advantageous Effects of Invention

According to the present disclosure, a work machine can be implemented for which the boom float function is adjustable depending on a current work mode.

BRIEF DESCRIPTION OF DRAWINGS

FIG. **1** is a perspective view schematically showing a configuration of a work machine according to one embodiment of the present disclosure.

FIG. **2** shows a configuration of a boom control system of the work machine shown in FIG. **1**.

FIG. **3** shows one example of functional blocks of the boom control system shown in FIG. **2**.

FIG. **4** is a flowchart showing one example of a boom control method for a work machine according to one embodiment of the present disclosure.

FIG. **5** shows respective states of a first valve and a second valve in a scrape-off mode.

FIG. **6** shows respective states of the first valve and the second valve in a breaker mode.

FIG. **7** shows respective states of the first valve and the second valve in an excavation assist mode.

FIG. **8** shows a relation (A) between the operation amount of boom lowering and the opening degree of the first valve and a relation (B) between the operation amount of boom lowering and the opening degree of the second valve, in the scrape-off mode.

FIG. **9** shows a relation (A) between the operation amount of boom lowering and the opening degree of the first valve and a relation (B) between the operation amount of boom lowering and the opening degree of the second valve, in the breaker mode.

FIG. **10** shows a relation (A) between the operation amount of boom lifting and the opening degree of the first valve and a relation (B) between the operation amount of boom lifting and the opening degree of the second valve, in the excavation assist mode.

FIG. **11** is a side view showing a configuration of a work machine including a breaker as an attachment.

DESCRIPTION OF EMBODIMENTS

Embodiments of the present disclosure are described hereinafter based on the drawings.

In the specification and the drawings, the same or corresponding components are denoted by the same reference characters, and a description thereof is not herein repeated. In the drawings, some components may not be shown or may be simplified for the sake of convenience of illustration. At least some of the embodiments and respective modifications may be combined with each other in any manner.

The present disclosure is applicable to work machines including at least a boom and a boom cylinder driving the boom, other than hydraulic excavator, and applicable to work machines manipulating the boom, such as wheel loader. In the following description, "upward," "downward," "front," "rear", "left" and "right" each refer to a direction with respect to an operator sitting on an operator's seat **4S** in an operator's cab **4** shown in FIG. **1**.

Configuration of Work Machine

First, a configuration of a work machine according to the present embodiment is described with reference to FIG. **1**.

FIG. **1** is a perspective view schematically showing a configuration of a work machine according to one embodiment of the present disclosure. As shown in FIG. **1**, hydraulic excavator **100** includes a main body **1** and a work implement **2** that is hydraulically actuated. Main body **1** includes a revolving unit **3** and a traveling unit **5**. Traveling unit **5** includes a pair of crawler belts **5**Cr and a travel motor **5**M. Rotation of crawler belts **5**Cr enables hydraulic excavator **100** to travel. Travel motor **5**M is provided as a driving source for traveling unit **5**. Travel motor **5**M is a hydraulic motor that is hydraulically actuated. Traveling unit **5** may include wheels (tires).

Revolving unit **3** is disposed on traveling unit **5** and supported by traveling unit **5**. Revolving unit **3** is enabled to revolve about a revolution axis RX, with respect to traveling unit **5**. Revolving unit **3** includes operator's cab **4**. In operator's cab **4**, operator's seat **4S** on which an operator is to sit is provided. An operator (driver) onboard operator's cab **4** can manipulate work implement **2**, manipulate revolving unit **3** to revolve with respect to traveling unit **5**, and manipulate hydraulic excavator **100** to travel by means of traveling unit **5**.

Revolving unit **3** includes an engine cover **9** and a counterweight provided in a rear part of revolving unit **3**. Engine cover **9** covers an engine compartment. In the engine compartment, an engine unit (engine, exhaust gas processing device, for example) are disposed.

Work implement **2** is supported by revolving unit **3**. Work implement **2** includes a boom **6**, a dipper stick **7**, and a bucket **8**. Work implement **2** further includes a boom cylinder **10**, a dipper stick cylinder **11**, and a bucket cylinder **12**.

Boom **6** is pivotally connected to main body **1** (traveling unit **5** and revolving unit **3**). Specifically, the proximal end of boom **6** is pivotally connected to revolving unit **3** with a boom hood pin **13** serving as a fulcrum.

Dipper stick **7** is pivotally connected to boom **6**. Specifically, the proximal end of dipper stick **7** is pivotally connected to the distal end of boom **6**, with a boom top pin **14** serving as a fulcrum. Bucket **8** is pivotally connected to dipper stick **7**. Specifically, the proximal end of bucket **8** is pivotally connected to the distal end of dipper stick **7** with a dipper stick top pin **15** serving as a fulcrum.

One end of boom cylinder **10** is connected to revolving unit **3** and the other end thereof is connected to boom **6**. Boom **6** can be driven by boom cylinder **10** with respect to main body **1**. Boom **6** can thus be driven to pivot upward/downward with respect to revolving unit **3**, with boom hood pin **13** serving as a fulcrum.

One end of dipper stick cylinder **11** is connected to boom **6** and the other end thereof is connected to dipper stick **7**. Dipper stick **7** can be driven with respect to boom **6** by dipper stick cylinder **11**. Dipper stick **7** can thus be driven to pivot upward/downward or fore/aft direction with respect to boom **6**, with boom top pin **14** serving as a fulcrum.

One end of bucket cylinder **12** is connected to dipper stick **7**, and the other end thereof is connected to a bucket link. Bucket **8** can be driven with respect to dipper stick **7** by bucket cylinder **12**. Bucket **8** can thus be driven to pivot upward/downward with respect to dipper stick **7**, with dipper stick top pin **15** serving as a fulcrum.

Configuration of Boom Control System

Next, a configuration of a boom control system according to the present embodiment is described with reference to FIG. **2**.

FIG. **2** shows a configuration of a boom control system of the work machine shown in FIG. **1**. As shown in FIG. **2**, the control system of boom **6** of work machine **100** includes a boom cylinder **10**, a hydraulic pump **20**, a first valve **21**, a second valve **22**, check valves **23**, **24**, an oil tank **25**, a controller (control unit) **30**, operation apparatuses **16***a* to **16***c*, a work mode setting unit **17**, and a float switch unit **18**.

Boom cylinder **10** includes a head side oil chamber **10***h* and a bottom side oil chamber **10***b*. Hydraulic pump **20** supplies hydraulic oil to each of head side oil chamber **10***h* and bottom side oil chamber **10***b* of boom cylinder **10**.

First valve **21** includes openings **21***a*, **21***b*, and first opening **21***c*. Opening **21***a* is a port connected to hydraulic pump **20**. Opening **21***b* is a port connected to head side oil chamber **10***h*. First opening **21***c* is connected to oil tank **25** to discharge hydraulic oil from head side oil chamber **10***h* to oil tank **25**.

Second valve **22** includes openings **22***a*, **22***b*, and second opening **22***c*. Opening **22***a* is a port connected to hydraulic pump **20**. Opening **22***b* is a port connected to bottom side oil chamber **10***b*. Second opening **22***c* is connected to oil tank **25** to discharge hydraulic oil from bottom side oil chamber **10***b* to oil tank **25**.

First valve **21** and second valve **22** each include a spool. The spool of first valve **21** and the spool of second valve **22** are designed to have the same dimensions.

First valve **21** is connected between head side oil chamber **10***h* and hydraulic pump **20**. Thus, hydraulic oil can be supplied from hydraulic pump **20** to head side oil chamber **10***h* through first valve **21**.

Oil tank **25** is connected to head side oil chamber **10***h* through first valve **21**. Thus, hydraulic oil in head side oil chamber **10***h* can be discharged to oil tank **25** through first valve **21**.

Head side oil chamber **10***h* is connected to oil tank **25** via check valve **23**. Thus, oil in oil tank **25** can be supplied into head side oil chamber **10***h* through check valve **23**.

Second valve **22** is connected between bottom side oil chamber **10***b* and hydraulic pump **20**. Thus, hydraulic oil can be supplied from hydraulic pump **20** to bottom side oil chamber **10***b* through second valve **22**.

Oil tank **25** is connected to bottom side oil chamber **10***b* through second valve **22**. Thus, hydraulic oil in bottom side oil chamber **10***b* can be discharged to oil tank **25** through second valve **22**.

Bottom side oil chamber **10***b* is connected to oil tank **25** through check valve **24**. Thus, oil in oil tank **25** can be supplied into bottom side oil chamber **10***b* through check valve **24**.

Operation apparatus **16***a* is a control lever, for example, for an operator to manipulate operation of boom **6**. Operation apparatus **16***b* is a control lever, for example, for an operator to manipulate operation of dipper stick **7**. Operation apparatus **16***c* is a control lever, for example, for an operator to manipulate operation of bucket **8**. Respective amounts of manipulation of operation apparatuses **16***a* to **16***c* are detected, for example, by a potentiometer, a hall IC (Integrated Circuit) or the like, and input as control signals to controller **30**.

Work mode setting unit **17** is an input device manipulated for input by an operator, for example. Work mode setting unit **17** may also be a display apparatus constituted of a touch panel, for example. In this case, a plurality of work modes of work implement **2** are displayed on work mode setting unit **17**. Work modes of work implement **2** are, for example, scrape-off mode, breaker mode, excavation assist mode, and the like. An operator selects and touches one of a plurality of work modes displayed on work mode setting unit **17**. A signal representing the work mode selected by the operator is input, as a control signal, to controller **30**.

Scrape-off work is work of scraping the ground surface for land grading. Breaker work is a work of breaking rocks or hard stratum.

Float switch unit **18** is a switch, for example. An operator can manipulate float switch unit **18** to selectively make a switch between execution and non-execution of the boom float function. A switch signal representing execution or non-execution selected by the operator is input, as a control signal, to controller **30**.

To controller **30**, respective control signals of operation apparatuses **16***a* to **16***c*, work mode setting unit **17**, and float switch unit **18** are input. During work including manipulation of boom **6**, controller **30** individually controls operation of respective spools of first valve **21** and second valve **22**, based on input control signals. Thus, during work including manipulation of boom **6**, the opening degree of first opening **21***c* and the opening degree of second opening **22***c* are controlled individually by controller **30**. Controller **30** individually controls respective opening degrees of first opening **21***c* and second opening **22***c*, based on the work mode selected through work mode setting unit **17**.

Configuration of Functional Blocks of Boom Control System

Next, a configuration of functional blocks of the boom control system shown in FIG. **2** is described with reference to FIG. **3**.

FIG. **3** shows one example of functional blocks of the boom control system shown in FIG. **2**. As shown in FIG. **3**, controller **30** includes a work mode determination unit **31**, a float switch determination unit **32**, a float operation start determination unit **33**, a first valve controller **34**, and a second valve controller **35**.

Work mode determination unit **31** receives, from work mode setting unit **17**, a control signal representing a work mode. Work mode determination unit **31** determines which work mode is selected by an operator, based on the control signal input from work mode setting unit **17**.

The work mode includes, for example, scrape-off mode, breaker mode, and excavation assist mode. The scrape-off mode is a mode of setting boom **6** in a state for causing bucket **8** to move along an uneven ground surface during scrape-off work. The breaker mode is a mode of setting for reducing vibrations of the work implement due to a breaker **8***a*, when breaker **8***a* is used as an attachment as shown in FIG. **11**. The excavation assist mode is a mode of setting boom **6** in a state for releasing load on bucket **8** during excavation.

Work mode determination unit **31** determines whether the work mode selected by an operator is the scrape-off mode, the breaker mode, or the excavation assist mode, for example. Work mode determination unit **31** outputs, to float switch determination unit **32**, a determination signal representing its determination.

Float switch determination unit **32** receives, from float switch unit **18**, a switch signal representing execution or non-execution of the boom float function. Receiving the determination signal from work mode determination unit **31**, float switch determination unit **32** determines which of execution and non-execution of the boom float function has been selected, based on the switch signal input from float switch unit **18**. Float switch determination unit **32** outputs, to float operation start determination unit **33**, a signal representing its determination.

Based on manipulation of operation apparatus **16***a* by an operator, float operation start determination unit **33** determines whether to start a boom float operation or not. For example, when the work mode is the scrape-off mode or the breaker mode, float operation start determination unit **33** determines that the boom float operation is to be started, based on a manipulation signal for lowering the boom. When the work mode is the excavation assist mode, for example, float operation start determination unit **33** determines that the boom float operation is to be started, based on the manipulation signal for lifting the boom.

When float operation start determination unit **33** determines that the boom float operation is to be started, float operation start determination unit **33** outputs a control signal base on the amount of manipulation of operation apparatus **16***a* to each of first valve controller **34** and second valve controller **35**. Thus, the boom float operation is started, as triggered by input of the manipulation signal from operation apparatus **16***a* to float operation start determination unit **33**.

Based on the control signal from float operation start determination unit **33**, first valve controller **34** controls operation of first valve **21**. Based on the control signal from float operation start determination unit **33**, second valve controller **35** controls operation of second valve **22**.

Boom Control Method

Next, a boom control method performed by the boom control system is described with reference to FIGS. **2** to **10**.

FIG. **4** is a flowchart showing one example of a boom control method for a work machine according to one embodiment of the present disclosure. FIGS. **5**, **6**, and **7** show respective states of the first valve and the second valve in the scrape-off mode, the breaker mode, and the excavation assist mode, respectively. FIGS. **8**, **9**, and **10** show a relation (A) between the amount of manipulation of the boom and the opening degree of the first valve and a relation (B) between the amount of manipulation of the boom and the opening degree of the second valve, in the scrape-off mode, the breaker mode, and the excavation assist mode, respectively.

As shown in FIGS. **3** and **4**, an operator inputs a work mode by means of work mode setting unit **17**. At this time, the operator touches any one of a plurality of work modes displayed on work mode setting unit **17**, for example. Thus, one work mode is selected through work mode setting unit **17**.

A plurality of work modes include, for example, scrape-off mode, breaker mode, and excavation assist mode, as described above. Through the operator's input, any one of the scrape-off mode, the breaker mode, and the excavation assist mode is selected, for example. A signal representing the work mode selected by the operator is input, as a control signal, to work mode determination unit **31** of controller **30** (step S**1**, FIG. **4**).

Receiving the control signal representing the work mode from work mode setting unit **17**, work mode determination unit **31** determines which work mode is selected by the operator, based on the control signal (step S**2**). For example, work mode determination unit **31** determines whether the work mode selected by the operator is the scrape-off mode, the breaker mode, or the excavation assist mode.

(Scrape-Off Mode)

When work mode determination unit **31** determines that the work mode is the scrape-off mode, float switch determination unit **32** determines whether the boom float function is active or not, based on the switch signal from float switch unit **18** (step S**3***a*, FIG. **4**).

When float switch determination unit **32** determines that the boom float function is non-active, normal control is performed (step S**4***a*, FIG. **4**). Under normal control, boom **6**, dipper stick **7**, and bucket **8** (FIG. **1**) are driven in accordance with the amount of manipulation of operation apparatuses **16***a* to **16***c* (FIG. **2**).

When float switch determination unit **32** determines that the boom float function is active, float operation start determination unit **33** determines whether to start the float operation or not. The determination of whether to start the float operation or not is made based on whether boom lowering operation has been performed by an operator or not (step S5*a*, FIG. **4**). Float operation start determination unit **33** determines whether boom lowering operation has been performed by an operator or not, based on a manipulation signal input from operation apparatus **16***a*.

When float operation start determination unit **33** determines that boom lowering operation has not been performed, normal control is performed (step S4*a*, FIG. **4**).

When float operation start determination unit **33** determines that boom lowering operation has been performed, first valve controller **34** controls operation of first valve **21** and second valve controller **35** controls operation of second valve **22**. Thus, as shown in FIG. **5**, respective operations of first valve **21** and second valve **22** are controlled to open first opening **21***c* of first valve **21** and second opening **22***c* of second valve **22** (step S6*a*, FIG. **4**).

First valve **21** includes a spool **21***s* that controls open/close of each of openings **21***a*, **21***b* and first opening **21***c*. Second valve **22** includes a spool **22***s* that controls open/close of each of openings **22***a*, **22***b* and second opening **22***c*. First valve **21** and second valve **22** each have a solenoid (not shown), for example.

Controller **30** inputs an electrical signal to the solenoid of first valve **21** to thereby control and drive spool **21***s* of first valve **21**. Thus, operation of spool **21***s* is controlled to open each of opening **21***b* and first opening **21***c* of first valve **21**. Thus, hydraulic oil in head side oil chamber **10***h* of boom cylinder **10** can be discharged to oil tank **25** through opening **21***b* and first opening **21***c*.

Controller **30** inputs an electrical signal to the solenoid of second valve **22** to thereby control and drive spool **22***s* of second valve **22**. Thus, operation of spool **22***s* is controlled to open each of opening **22***b* and second opening **22***c* of second valve **22**. Thus, hydraulic oil in bottom side oil chamber **10***b* of boom cylinder **10** can be discharged to oil tank **25** through opening **22***b* and second opening **22***c*.

As shown in FIG. **5**, when boom cylinder **10** is extended in the scrape-off mode, hydraulic oil in head side oil chamber **10***h* is discharged to oil tank **25** through first valve **21**, and oil in oil tank **25** is supplied to bottom side oil chamber **10***b* through check valve **24**. When boom cylinder **10** is retracted in the scrape-off mode, hydraulic oil in bottom side oil chamber **10***b* is discharged to oil tank **25** through second valve **22**, and oil in oil tank **25** is supplied to head side oil chamber **10***h* through check valve **23**.

In the scrape-off mode, respective opening degrees D**1**, D**2** of first opening **21***c* and second opening **22***c* are controlled in accordance with the operation amount of lowering of boom **6** by operation apparatus **16***a*. Specifically, as shown in FIG. **8** (A), as the operation amount of lowering of boom **6** by operation apparatus **16***a* increases, controller **30** controls first valve **21** to make opening degree D**1** of first opening **21***c* larger. As shown in FIG. **8** (B), as the operation amount of lowering of boom **6** by operation apparatus **16***a* increases, controller **30** controls second valve **22** to make opening degree D**2** of second opening **22***c* larger.

As shown in FIG. **8** (A) and FIG. **8** (B), opening degree D**1** of first opening **21***c* may be substantially identical to opening degree D**2** of second opening **22***c*, with respect to the operation amount of lowering of the boom.

(Breaker Mode)

As shown in FIGS. **3** and **4**, when work mode determination unit **31** determines that the work mode is the breaker mode, float switch determination unit **32** determines whether the boom float function is active or not (step S3*b*, FIG. **4**). Float switch determination unit **32** determines whether the boom float function is active or not, based on the switch signal from float switch unit **18**.

When float switch determination unit **32** determines that the boom float function is non-active, normal control is performed (step S4*b*, FIG. **4**).

When float switch determination unit **32** determines that the boom float function is active, float operation start determination unit **33** determines whether to start the float operation or not. The determination of whether to start the float operation or not is made based on whether boom lowering operation has been performed by an operator or not (step S5*b*, FIG. **4**). Float operation start determination unit **33** determines whether boom lowering operation has been performed by an operator or not, based on a manipulation signal input from operation apparatus **16***a*.

When float operation start determination unit **33** determines that boom lowering operation has not been performed, normal control is performed (step S4*b*, FIG. **4**).

When float operation start determination unit **33** determines that boom lowering operation has been performed, first valve controller **34** controls operation of first valve **21** and second valve controller **35** controls operation of second valve **22**. Thus, as shown in FIG. **6**, respective operations of first valve **21** and second valve **22** are controlled to make opening degree D**1** of first opening **21***c* of first valve **21** smaller than opening degree D**2** of second opening **22***c* of second valve **22** (step S6*b*, FIG. **4**).

In the breaker mode, controller **30** controls spool **21***s* to close opening **21***a* and open opening **21***b* and first opening **21***c*. Thus, hydraulic oil in head side oil chamber **10***h* of boom cylinder **10** can be discharged to oil tank **25** through opening **21***b* and first opening **21***c*.

In the breaker mode, controller **30** also controls spool **22***s* to close opening **22***a* and open opening **22***b* and second opening **22***c*. Thus, hydraulic oil in bottom side oil chamber **10***b* of boom cylinder **10** can be discharged to oil tank **25** through opening **22***b* and second opening **22***c*.

Controller **30** performs control to make opening degree D**1** of first opening **21***c* smaller than opening degree D**2** of second opening **22***c*. Therefore, as shown in FIG. **9** (A) and FIG. **9** (B), as the operation amount of boom lowering increases, both of opening degrees D**1**, D**2** increase, while the rate of increase of opening degree D**1** is smaller than the rate of increase of opening degree D**2**.

In the breaker mode, first opening **21***c* may be closed completely by spool **21***s*.

As shown in FIG. **6**, when boom cylinder **10** is extended in the breaker mode, hydraulic oil in head side oil chamber **10***h* is discharged to oil tank **25** through first valve **21**, and oil in oil tank **25** is supplied to bottom side oil chamber **10***b* through check valve **24**. When boom cylinder **10** is retracted in the breaker mode, hydraulic oil in bottom side oil chamber **10***b* is discharged to oil tank **25** through second valve **22**, and oil in oil tank **25** is supplied to head side oil chamber **10***h* through check valve **23**.

(Excavation Assist Mode)

As shown in FIGS. **3** and **4**, when work mode determination unit **31** determines that the work mode is the excavation assist mode, float switch determination unit **32** determines whether the boom float function is active or not (step S3*c*, FIG. **4**). Float switch determination unit **32** determines whether the boom float function is active or not, based on the switch signal from float switch unit **18**.

When float switch determination unit **32** determines that the boom float function is non-active, normal control is performed (step S4*c*, FIG. **4**).

When float switch determination unit **32** determines that the boom float function is active, float operation start determination unit **33** determines whether to start the float operation or not. The determination of whether to start the float operation or not is made based on whether boom lifting operation has been performed by an operator or not (step S5*c*, FIG. **4**). Float operation start determination unit **33** determines whether boom lifting operation has been performed by an operator or not, based on a manipulation signal input from operation apparatus **16***a*.

When float operation start determination unit **33** determines that boom lifting operation has not been performed, normal control is performed (step S4*c*, FIG. **4**).

When float operation start determination unit **33** determines that boom lifting operation has been performed, first valve controller **34** controls operation of first valve **21** and second valve controller **35** controls operation of second valve **22**. Thus, as shown in FIG. **7**, control is performed to make opening degree D**2** of second opening **22***c* smaller than opening degree D**1** of first opening **21***c* (step S6*c*, FIG. **4**). At this time, second opening **22***c* is closed, for example.

In the excavation assist mode, controller **30** controls spool **21***s* to open opening **21***b* and first opening **21***c*. Thus, hydraulic oil in head side oil chamber **10***h* of boom cylinder **10** can be discharged to oil tank **25** through opening **21***b* and first opening **21***c*.

In the excavation assist mode, opening degree D**2** of second opening **22***c* is made smaller than opening degree D**1** of first opening **21***c*, or set to zero (closed). Preferably, second opening **22***c* is completely closed.

In the excavation assist mode, respective opening degrees of first opening **21***c* and second opening **22***c* are controlled in accordance with the operation amount of lifting of boom **6** by operation apparatus **16***a*. Specifically, as shown in FIG. **10** (A), as the operation amount of lifting of boom **6** by operation apparatus **16***a* increases, controller **30** controls first valve **21** to make opening degree D**1** of first opening **21***c* larger. As shown in FIG. **10** (B), even when the operation amount of lifting of boom **6** by operation apparatus **16***a* increases, opening degree D**2** of second opening **22***c* does not substantially increase, or second opening **22***c* remains closed.

As seen from the foregoing, in the present embodiment, controller **30** individually controls opening degree D**1** of first opening **21***c* of first valve **21** and opening degree D**2** of second opening **22***c* of second valve **22**. Controller **30** also individually controls opening degrees D**1**, D**2** based on the work mode (scrape-off mode, breaker mode, excavation assist mode, for example) of work implement **2**.

Based on the result of determination that the work mode is the scrape-off mode, controller **30** performs control to open both first opening **21***c* and second opening **22***c*, as shown in FIGS. **5** and (A) and (B) of FIG. **8**. Based on the result of determination that the work mode is the breaker mode, controller **30** performs control to make opening degree D**1** smaller than opening degree D**2** as shown in FIGS. **6** and (A) and (B) of FIG. **9**. Based on the result of determination that the work mode is the excavation assist mode, controller **30** performs control to make opening degree D**2** smaller than opening degree D**1** as shown in FIGS. **7** and (A) and (B) of FIG. **10**.

Advantageous Effects

Next, advantageous effects of the present embodiment are described.

In the present embodiment, as shown in FIGS. **5** to **7**, controller **30** individually controls opening degree D**1** of first opening **21***c* of first valve **21** and opening degree D**2** of second opening **22***c* of second valve **22**. Thus, hydraulic oil in head side oil chamber **10***h* of boom cylinder **10** and hydraulic oil in bottom side oil chamber **10***b* thereof can be controlled individually to be discharged to oil tank **25**. Thus, the boom float function can be adjusted depending on a current work mode, without providing an opening in the spool for boom float, or without preparing a valve for switching the boom float function, besides the main valve.

In the present embodiment, controller **30** individually controls opening degrees D**1**, D**2**, based on the work mode (scrape-off mode, breaker mode, excavation assist mode, for example) of work implement **2**. Thus, a work machine for which the boom float function is adjustable depending on a current work mode such as scrape-off mode, breaker mode, and excavation assist mode can be implemented.

In the present embodiment, as shown in FIGS. **5** and (A) and (B) of FIG. **8**, controller **30** performs control to open both first opening **21***c* and second opening **22***c*, based on the result of determination that the work mode is the scrape-off mode.

Thus, boom **6** is enabled to move freely upward/downward by external force. Therefore, during the scrape-off work, bucket **8** can be moved along an uneven ground surface easily. The boom float function can also be performed while bucket **8** is located in the air, so that boom **6** can be lowered to reach the ground by the weight of work implement **2**.

As shown in FIG. **11**, when breaker **8***a* is used as an attachment and a chisel Baa is struck in the air, a retainer pin and chisel Baa for example are likely to be chipped off and the distal end of a chisel holder is likely to be damaged. It is therefore necessary that the distal end of chisel Baa be in contact with an object to be crushed, when breaker **8***a* is caused to operate.

In the present embodiment, as shown in FIGS. **6** and (A) and (B) of FIG. **9**, controller **30** performs control to make opening degree D**1** smaller than opening degree D**2**, based on the result of determination that the work mode is the breaker mode.

Thus, while boom **6** is moved downward more easily (in the direction in which boom cylinder **10** is retracted), boom **6** is moved upward less easily (in the direction in which boom cylinder **10** is extended). Thus, in the breaker mode, the distal end of chisel Baa is prevented from being separated from an object to be crushed, so that chipping off of the retainer pin and chisel Baa for example, and damage to the chisel holder, for example, can be prevented.

In the case of demolition or the like, hard objects may be crushed. In this case, there is a possibility that work implement **2** is broken, unless the load resultant from crushing is released.

In the present embodiment, as shown in FIGS. **7** and (A) and (B) of FIG. **10**, controller **30** performs control to make opening degree D**2** smaller than opening degree D**1**, based on the result of determination that the work mode is the excavation assist mode.

Thus, while boom **6** is moved downward less easily (in the direction in which boom cylinder **10** is retracted), boom **6** is moved upward more easily (in the direction in which boom cylinder **10** is extended). Thus, when a hard object is excavated, boom **6** can be moved away upward to release the load resultant from excavation. The durability of work implement **2** can be improved in this way.

Controller **30** shown in each of FIGS. **2** and **3** in the above-described embodiment may be mounted on work machine **100** or located away from work machine **100**. When controller **30** is located away from work machine **100**, controller **30** may be connected wirelessly to work mode setting unit **17**, float switch unit **18**, operation apparatuses **16***a* to **16***c*, first valve **21**, and second valve **22**, for example. Controller **30** may for example be a processor, and may be a CPU (Central Processing Unit).

It should be construed that the embodiments disclosed herein are given by way of illustration in all respects, not by way of limitation. It is intended that the scope of the present invention is defined by claims, not by the description above, and encompasses all modifications equivalent in meaning and scope to the claims.

REFERENCE SIGNS LIST

1 main body; **2** work implement; **3** revolving unit; **4** operator's cab; **4**S operator's seat; **5** traveling unit; **5**Cr crawler belt; **5**M travel motor; **6** boom; **7** dipper stick; **8** bucket; **8***a* breaker; **8***aa* chisel; **9** engine cover; **10** boom cylinder; **10***b* bottom side oil chamber; **10***h* head side oil chamber; **11** dipper stick cylinder; **12** bucket cylinder; **13** boom hood pin; **14** boom top pin; **15** arm top pin; **16***a*, **16***b*, **16***c* operation apparatus; **17** work mode setting unit; **18** float switch unit; **20** hydraulic pump; **21** first valve; **21***a*, **21***b*, **22***a*, **22***b* opening; **21***c* first opening; **21***s*, **22***s* spool; **22** second valve; **22***c* second opening; **23**, **24** check valve; **25** oil tank; **30**, **50** controller; **31** work mode determination unit; **32** float switch determination unit; **33** float operation start determination unit; **34** first valve controller; **35** second valve controller; **50***k* storage unit; **100** hydraulic excavator; D**1**, D**2** opening degree; RX revolution axis

The invention claimed is:

1. A boom control system of a work machine comprising a boom, the boom control system comprising:

a boom cylinder that drives the boom and includes a head side oil chamber and a bottom side oil chamber;

a hydraulic pump;

an oil tank;

a first valve that supplies oil from the hydraulic pump to the head side oil chamber and discharges the oil to the oil tank; and

a second valve that supplies oil from the hydraulic pump to the bottom side oil chamber and discharges the oil to the oil tank, wherein

the first valve includes a first opening for discharging the oil in the head side oil chamber to the oil tank,

the second valve includes a second opening for discharging the oil in the bottom side oil chamber to the oil tank, and

the boom control system further comprises a controller that individually controls

an opening degree of the first opening and an opening degree of the second opening, during work including manipulation of the boom and

upon activation of a boom float function adjusts at least one of the opening degree of the first opening and the opening degree of the second opening based on a work mode selected from a plurality of work modes of a work implement including the boom and the boom cylinder.

2. The boom control system of the work machine according to claim **1**, wherein the controller performs control to open both the first opening and the second opening, based on a result of determination that the work mode is a scrape-off mode.

3. The boom control system of the work machine according to claim **1**, wherein the controller performs control to make the opening degree of the first opening smaller than the opening degree of the second opening, based on a result of determination that the work mode is a breaker mode.

4. The boom control system of the work machine according to claim **1**, wherein the controller performs control to make the opening degree of the second opening smaller than the opening degree of the first opening, based on a result of determination that the work mode is an excavation assist mode.

5. The boom control system of the work machine according to claim **1**, wherein the controller is configured to adjust the boom float function based at least in part on a signal to activate the boom float function.

6. A boom control system of a work machine comprising a boom, the boom control system comprising:

a boom cylinder that drives the boom and includes a head side oil chamber and a bottom side oil chamber;

a hydraulic pump;

an oil tank;

a first valve that supplies oil from the hydraulic pump to the head side oil chamber and discharges the oil to the oil tank; and

a second valve that supplies oil from the hydraulic pump to the bottom side oil chamber and discharges the oil to the oil tank, wherein

the first valve includes a first opening for discharging the oil in the head side oil chamber to the oil tank,

the second valve includes a second opening for discharging the oil in the bottom side oil chamber to the oil tank, and

the boom control system further comprises a controller that individually controls a first opening degree of the first opening and a second opening degree of the second opening, during work including manipulation of the boom,

wherein the controller individually controls the first opening degree of the first opening and the second opening degree of the second opening, based on a work mode of a work implement including the boom and the boom cylinder,

in response to a result of determination that the work mode is a scrape-off mode, the controller performs control to open both the first opening and the second opening,

in response to a result of determination that the work mode is a breaker mode, the controller performs control to make the first opening degree of the first opening smaller than the second opening degree of the second opening, and

in response to a result of determination that the work mode is an excavation assist mode, the controller performs control to make the second opening degree of the second opening smaller than the first opening degree of the first opening.

7. A boom control system of a work machine, the boom control system comprising:

a boom;

a boom cylinder that drives the boom and includes a head side oil chamber and a bottom side oil chamber;

a hydraulic pump;

an oil tank;

a first valve that supplies oil from the hydraulic pump to the head side oil chamber and discharges the oil to the oil tank; and

a second valve that supplies oil from the hydraulic pump to the bottom side oil chamber and discharges the oil to the oil tank, wherein

the first valve includes a first opening for discharging the oil in the head side oil chamber to the oil tank,

the second valve includes a second opening for discharging the oil in the bottom side oil chamber to the oil tank, and

the boom control system further comprises a controller that individually controls an opening degree of the first opening and an opening degree of the second opening upon activation of a boom float function and based on a result of determination of a type of a work mode of a work implement having the boom and the boom cylinder, during work including manipulation of the boom.

8. The boom control system of the work machine according to claim **7**, wherein the controller performs control to open both the first opening and the second opening, based on the result of determination that the work mode is a scrape-off mode.

9. The boom control system of the work machine according to claim **7**, wherein the controller performs control to make the opening degree of the first opening smaller than the opening degree of the second opening, based on the result of determination that the work mode is a breaker mode.

10. The boom control system of the work machine according to claim **7**, wherein the controller performs control to make the opening degree of the second opening smaller than the opening degree of the first opening, based on the result of determination that the work mode is an excavation assist mode.

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